# AIO in Postgres 18 and beyond

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https://anarazel.de/talks/2025-10-23-pgconf-eu-aio-in-PG-18-and-beyond/aio-in-PG-18-and-beyond.pdf

Microsoft

# Thanks

- A lot of work by a lot of folks
- Thomas, Melanie, Bilal, Noah, Heikki, Robert, …
- Microsoft
- Tomas and Peter G. for index prefetching work
- Lots of others

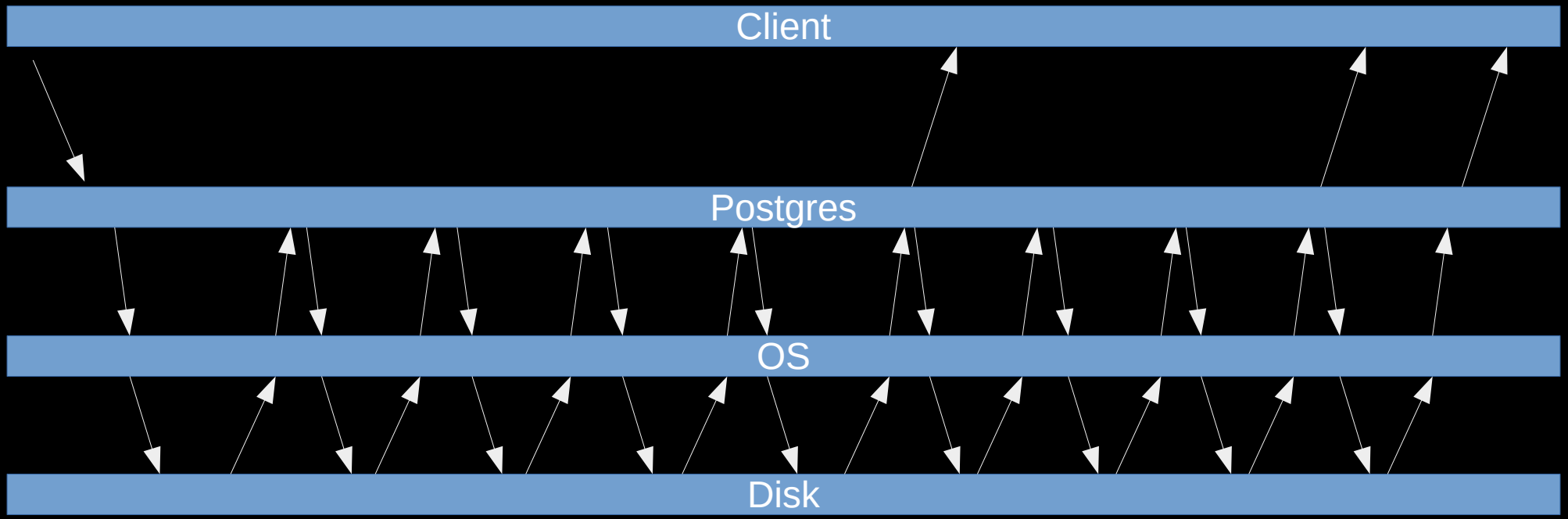
Microsoft

# AIO?

Asynchronous Input/Output

Microsoft

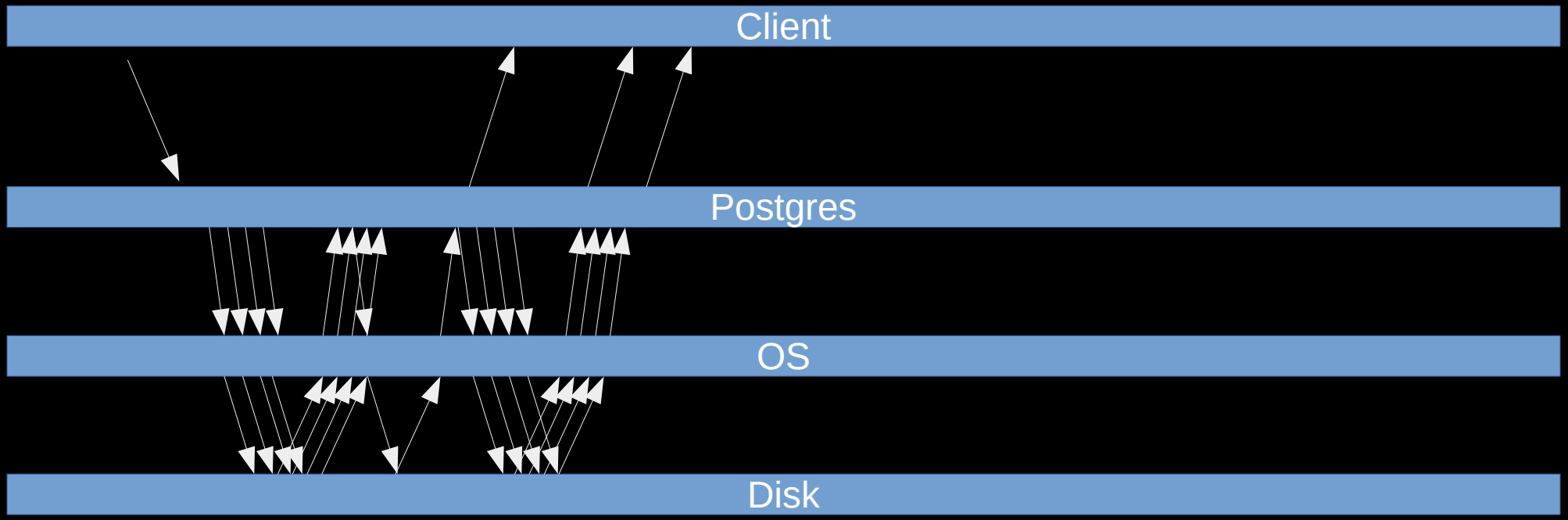
# synchronous, not cached




Time

Microsoft

# asynchronous, not cached



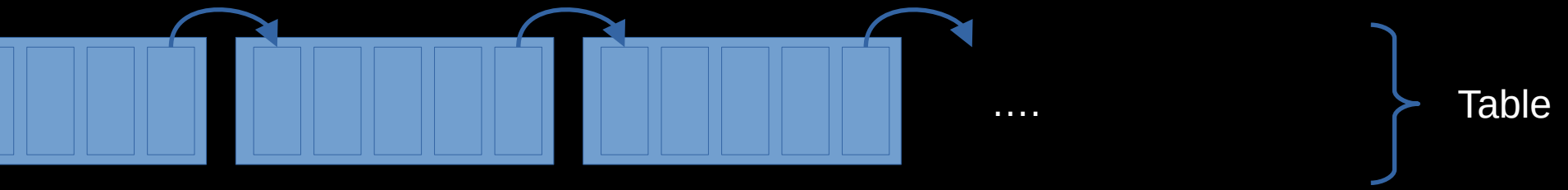

Time

Microsoft

# Predict the Future

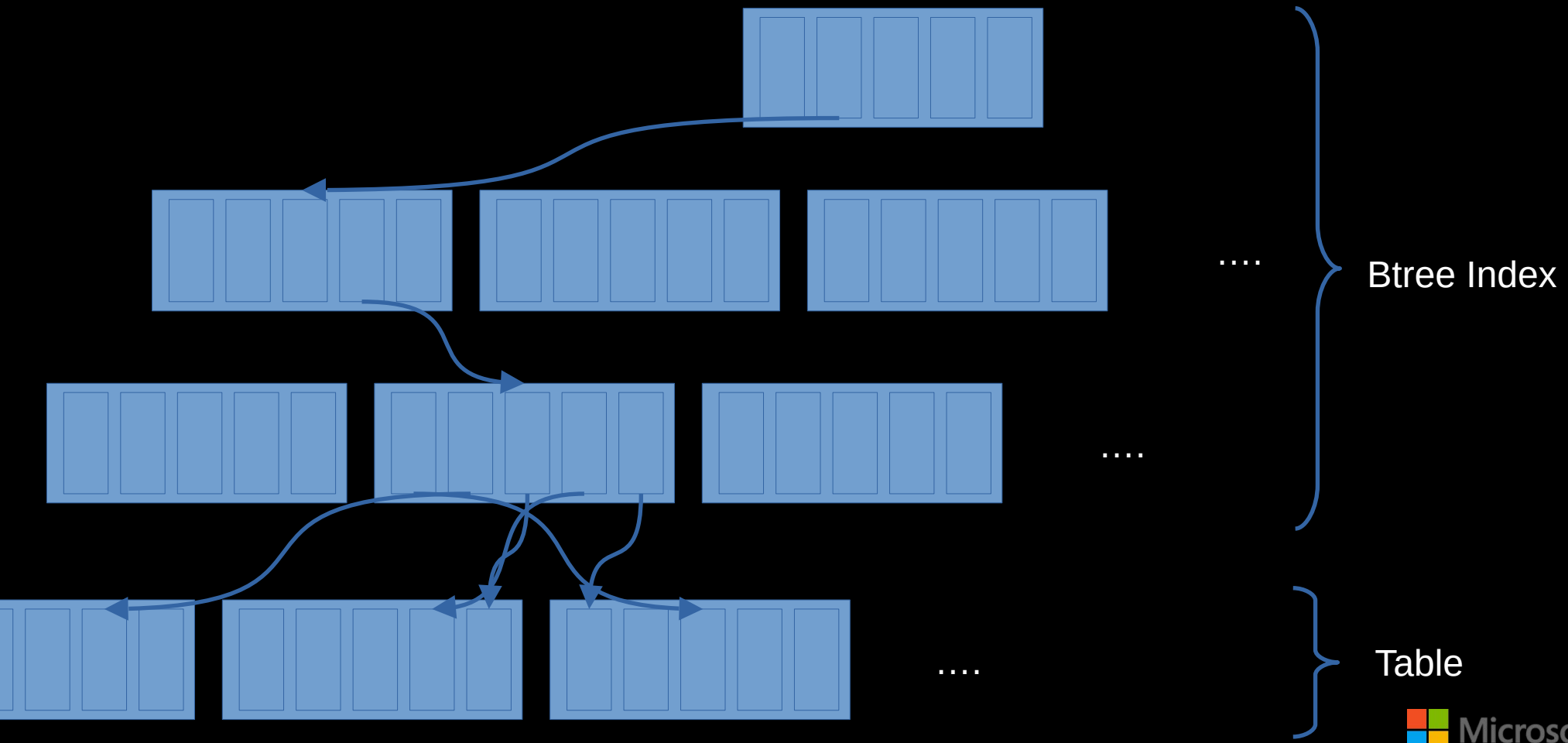
Microsoft

# Sequential Scan




Microsoft

# Index Scan




Microsoft

16:
- Buffer Manager Infra
- Relation  Extension

17:
- Read Streams
- Streamify
    - Seq Scan
    - Analyze
    - Prewarm
- Experimental Direct I/O

18:
- AIO Infra
- AIO for buffered reads
- Streamify
    - Bitmap Heap Scan
    - Vacuum
    - autoprewarm
    - CREATE DATABASE
    - amcheck

Microsoft

# 18: io_method = worker

- portable
- parallelizes checksums, memory copy
- limited I/O depth, particularly with high latency storage
- global
- number of workers controlled by io_workers

Microsoft

# 18: io_method = io_uring

- linux specific, better with recent-ish kernels
- lower latency
- deep I/O queues
- per backend
- does **not** parallelize checksum computation
- requires tuning of file descriptor limits

Microsoft

# 18: io_method = sync

- don’t use AIO
- behaves as close as realistic to < 18
- “safety net”

Microsoft

# 18: When can AIO help?

- IO bound
  - track_io_timing
  - EXPLAIN (ANALYZE, BUFFERS)
- only for reads
- foreground: seqscan, bitmap heap scan
- background: vacuum
- Just the absolute basics!

Microsoft

# Benchmark Setup

- 2x Gold 6442Y, 256GB RAM
- 2x Samsung SSD 2TB PM9A1, striped, XFS
- Linux 6.17
- Optional: Artificial 1ms latency added with dm_delay
- io_workers=32, effective_io_concurrency=32, shared_buffers=32GB
- checksums enabled on all branches
- 16 has support for cache clearing added

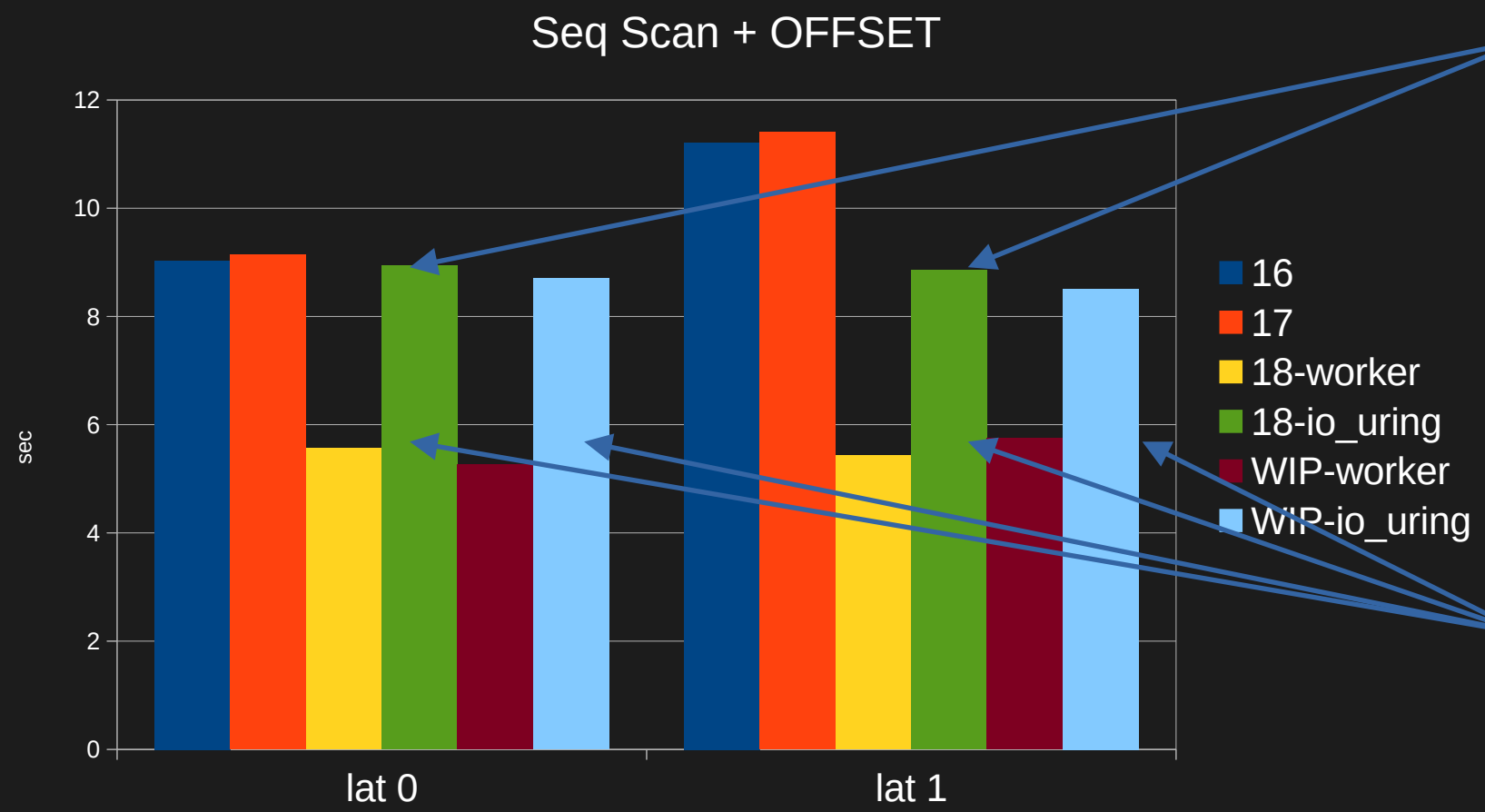
Microsoft

# Benchmark Workload

- Large table with sequential and random columns
- Table populated in parallel
- PG & OS cache is cleared between queries

Microsoft

# 18: AIO for Seq Scans

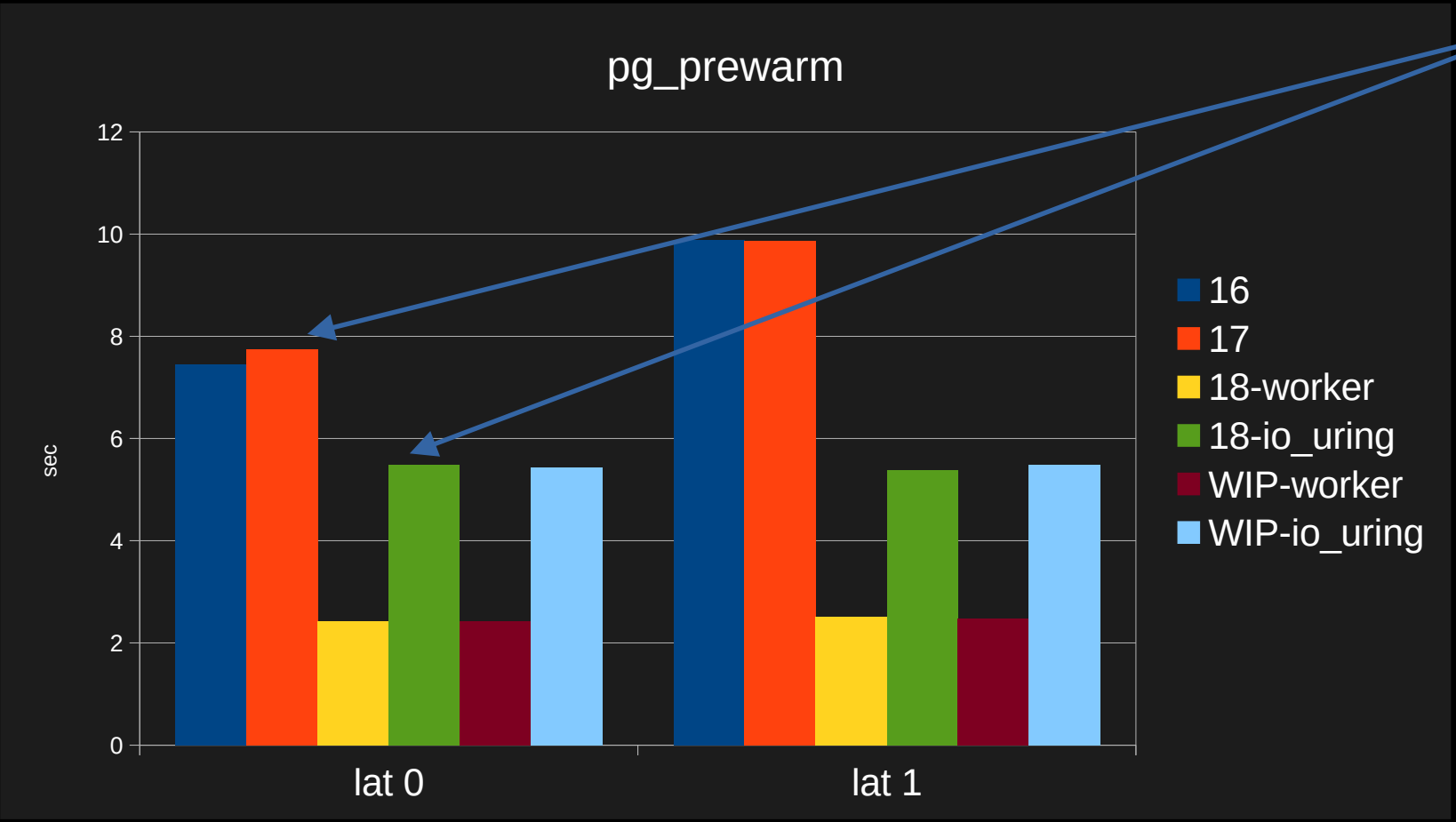



- Latency has little effect with AIO
- CPU bottlenecked (query + checksums)

Checksums not parallelized

Microsoft

# 18: AIO for prewarm




- bigger difference without checksum

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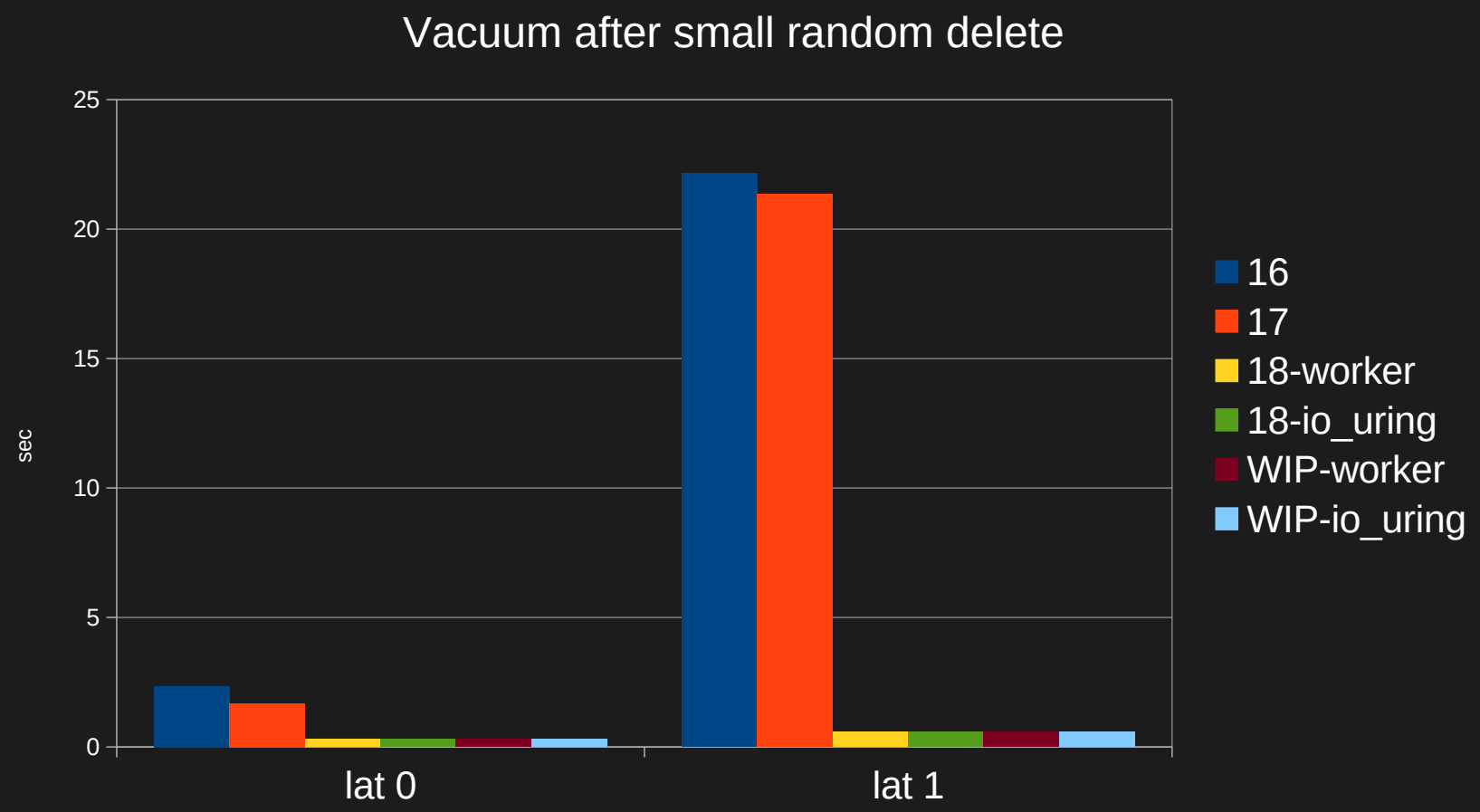
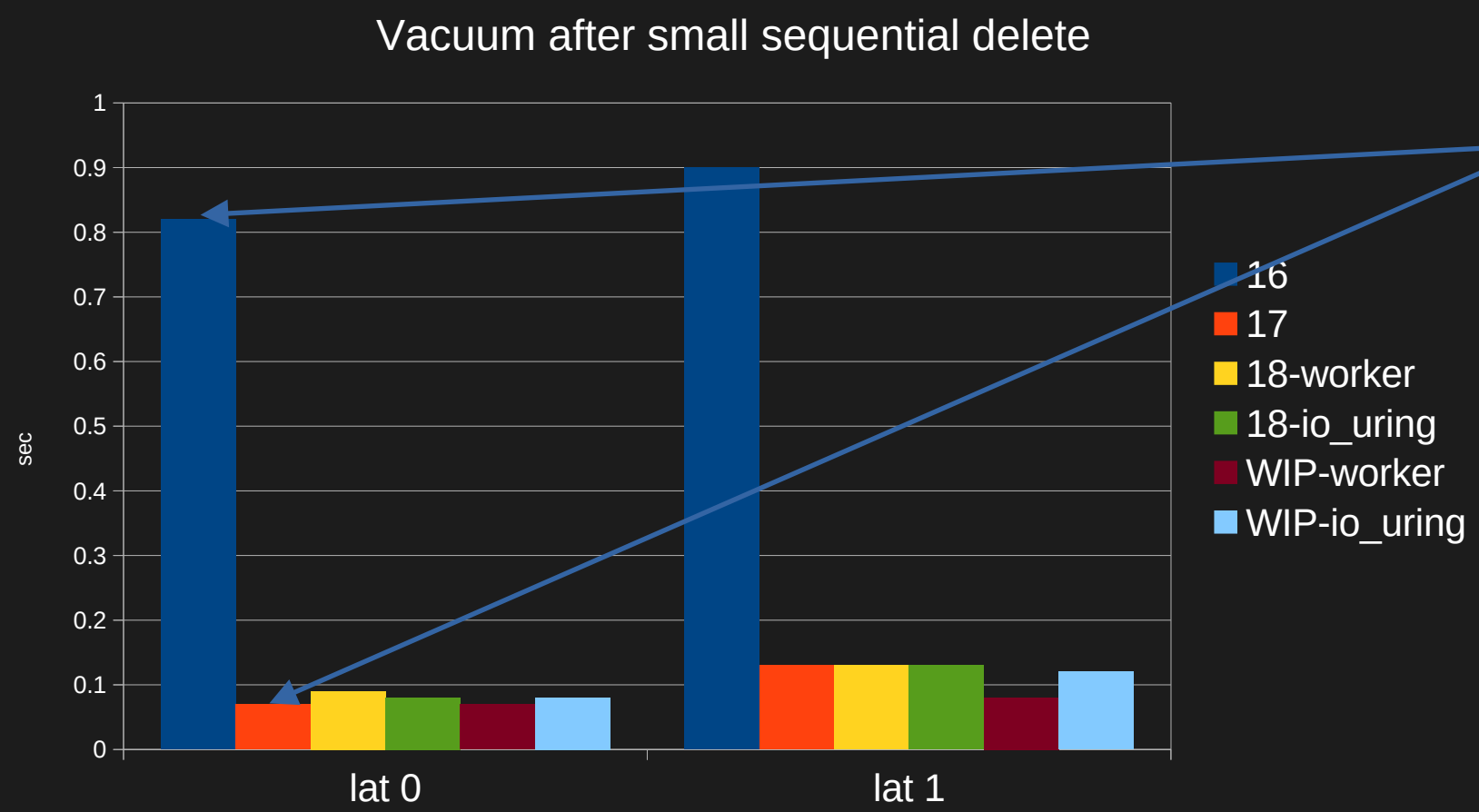
# 18: AIO for Vacuum




- huge effect
- only if small portion of table changed
  - can be generalized

Microsoft

# 18: AIO for Vacuum




- AIO has no effect → OS readahead
- CPU efficiency improvement in 17

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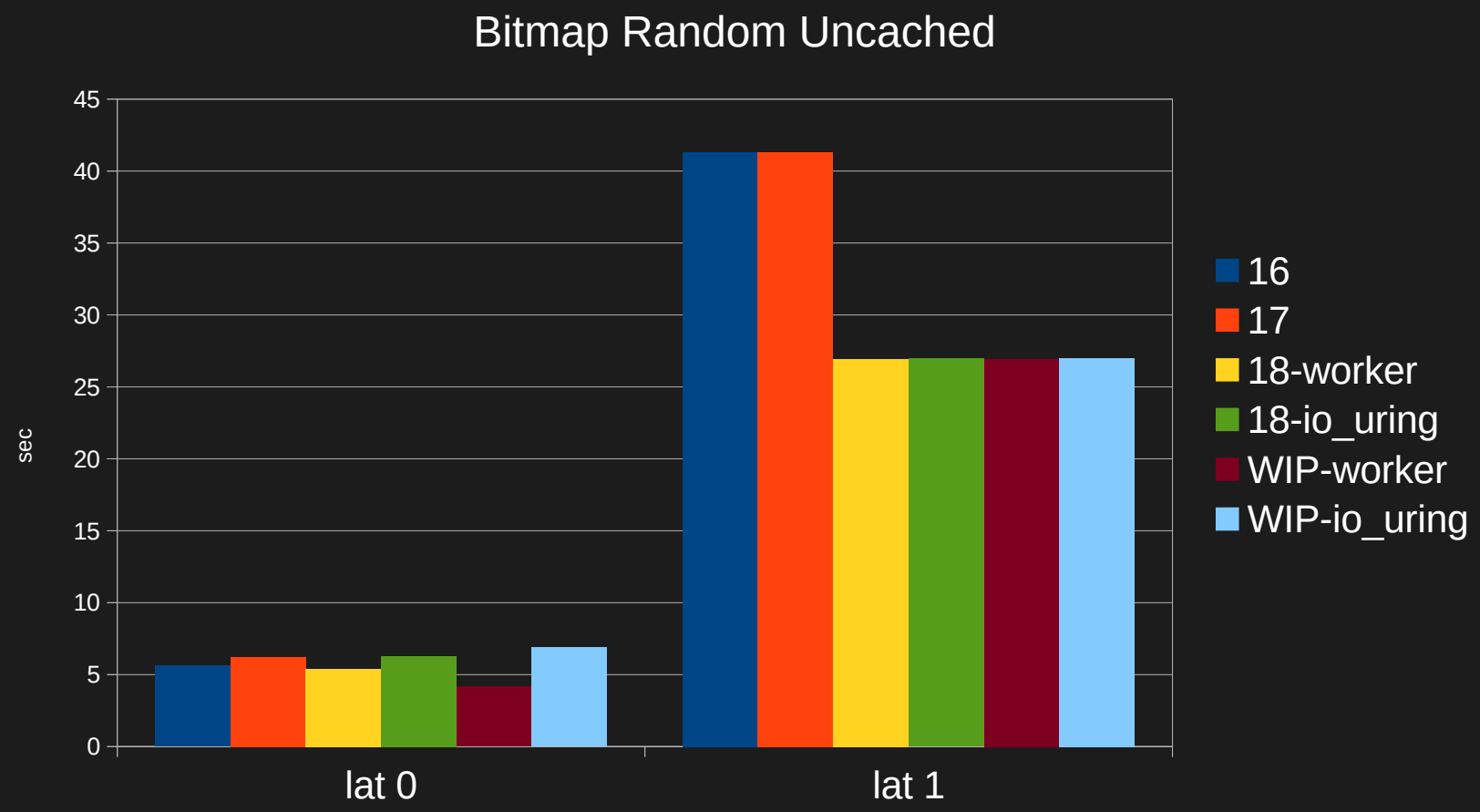
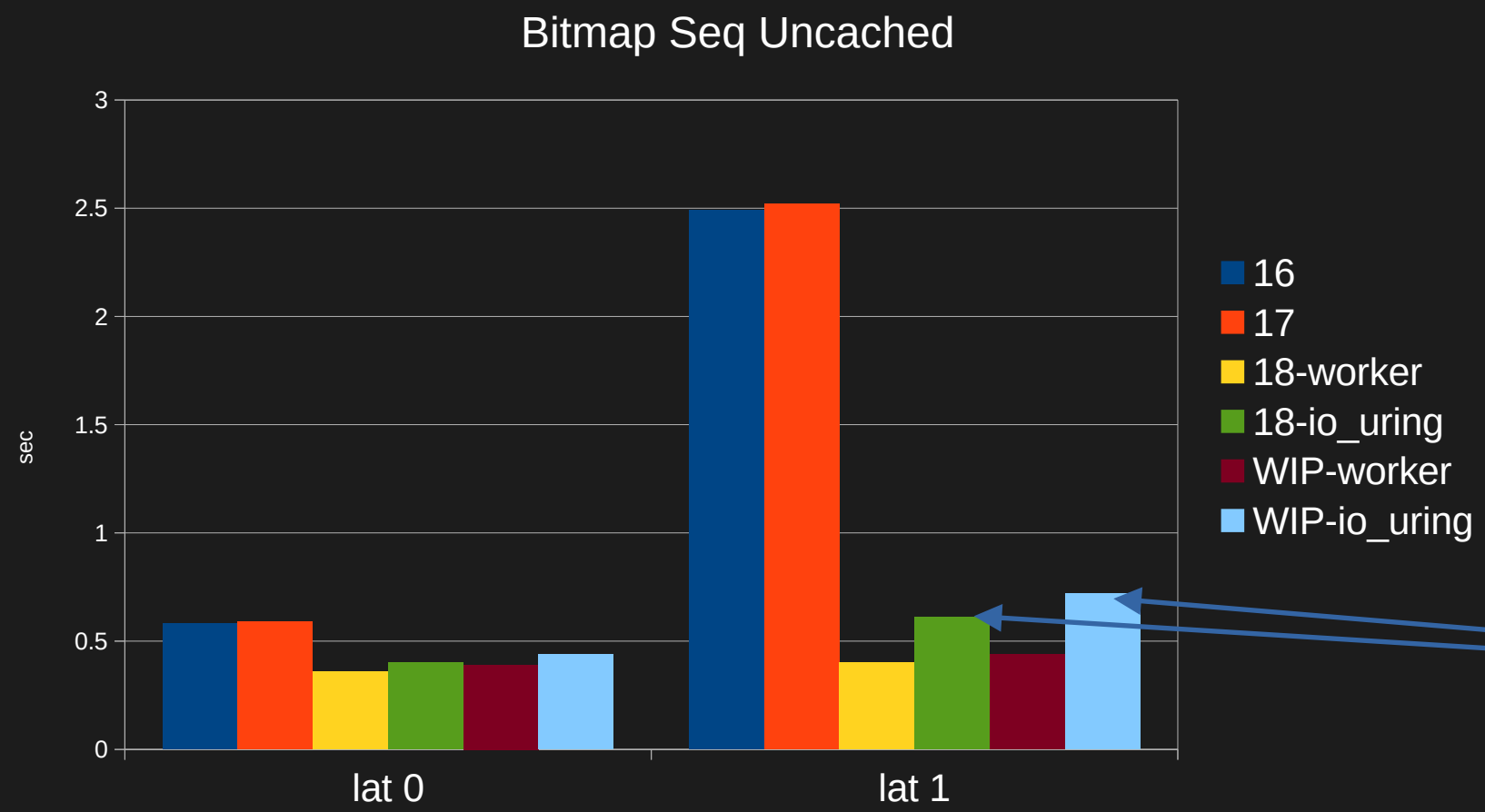
# 18: AIO for Bitmap Heap Scan




- already did prefetching
- but now with DIO
- more precise control of IO depth

Microsoft

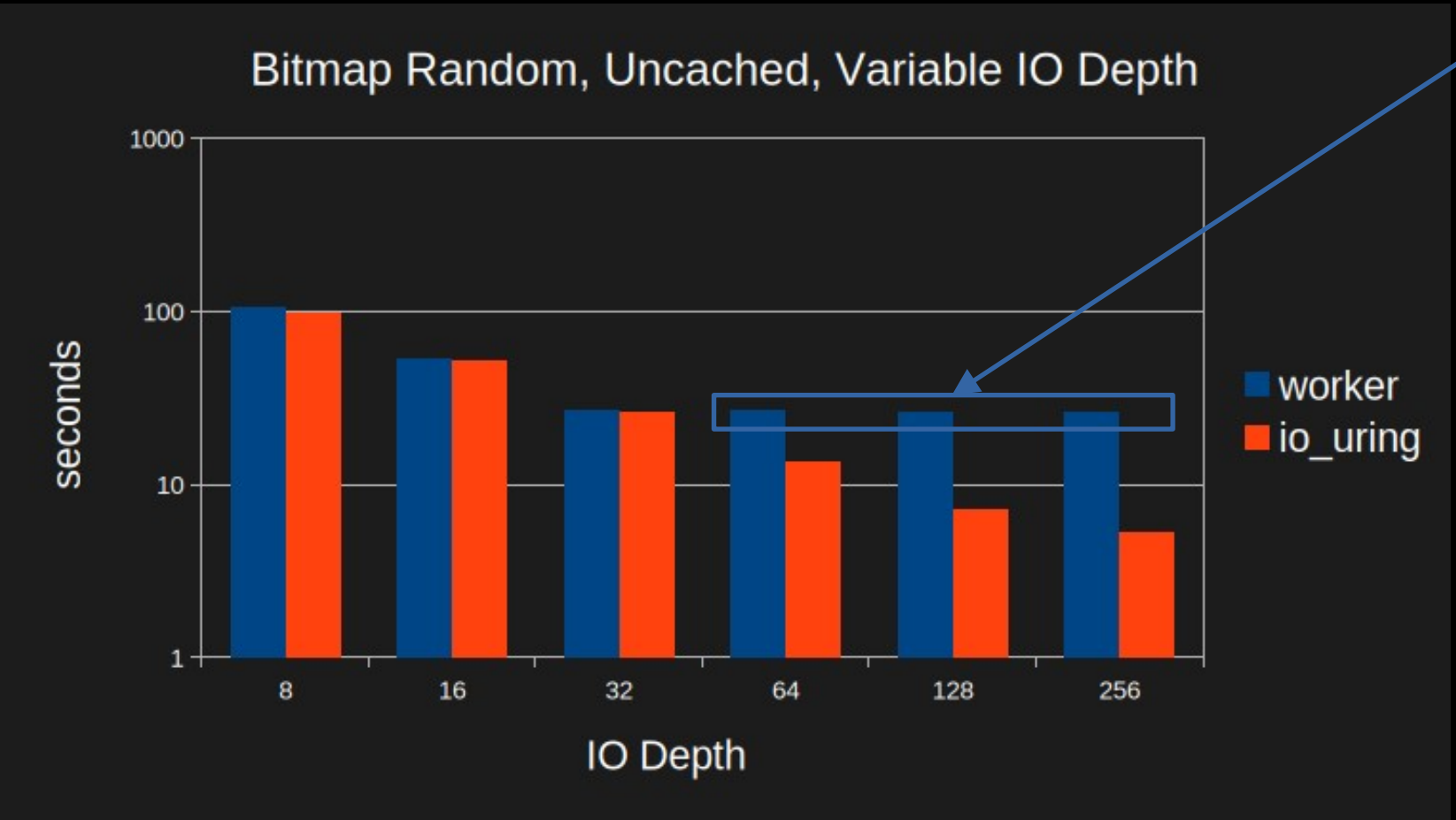
# 18: AIO for Bitmap Heap Scan




Checksums
not parallelized

Microsoft

# IO Depth vs io_method




- number of workers limits IO depth

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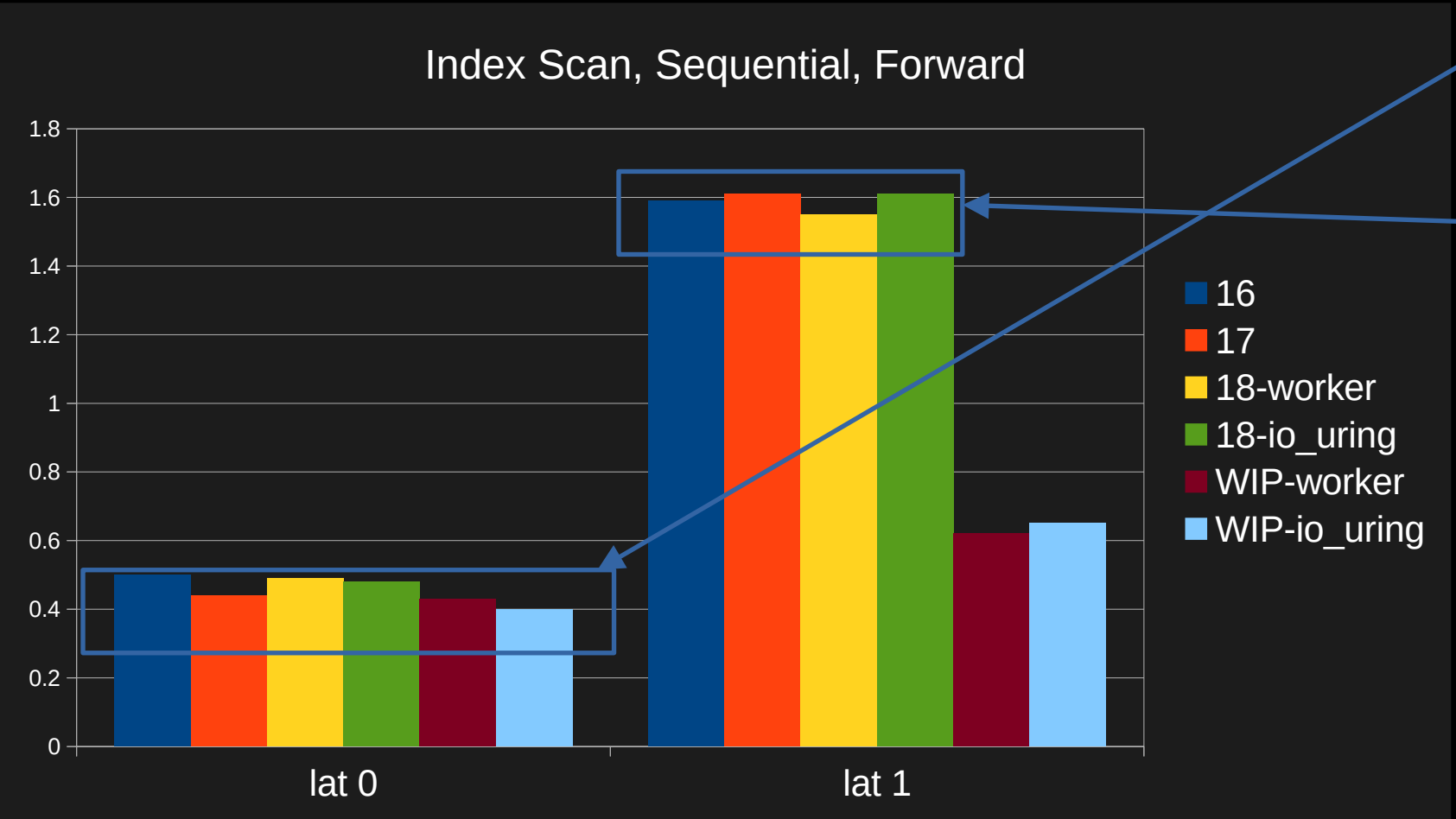
# 19? 20?: Index Readahead

- Tomas Vondra w/ help from Peter Geoghegan
- Much harder than already-existing AIO users
- Other performance benefits plausible
- Some regression potential too

Microsoft

# 19? 20?: Index Readahead




- kernel readahead suffices on low-delay storage
- No PG readahead

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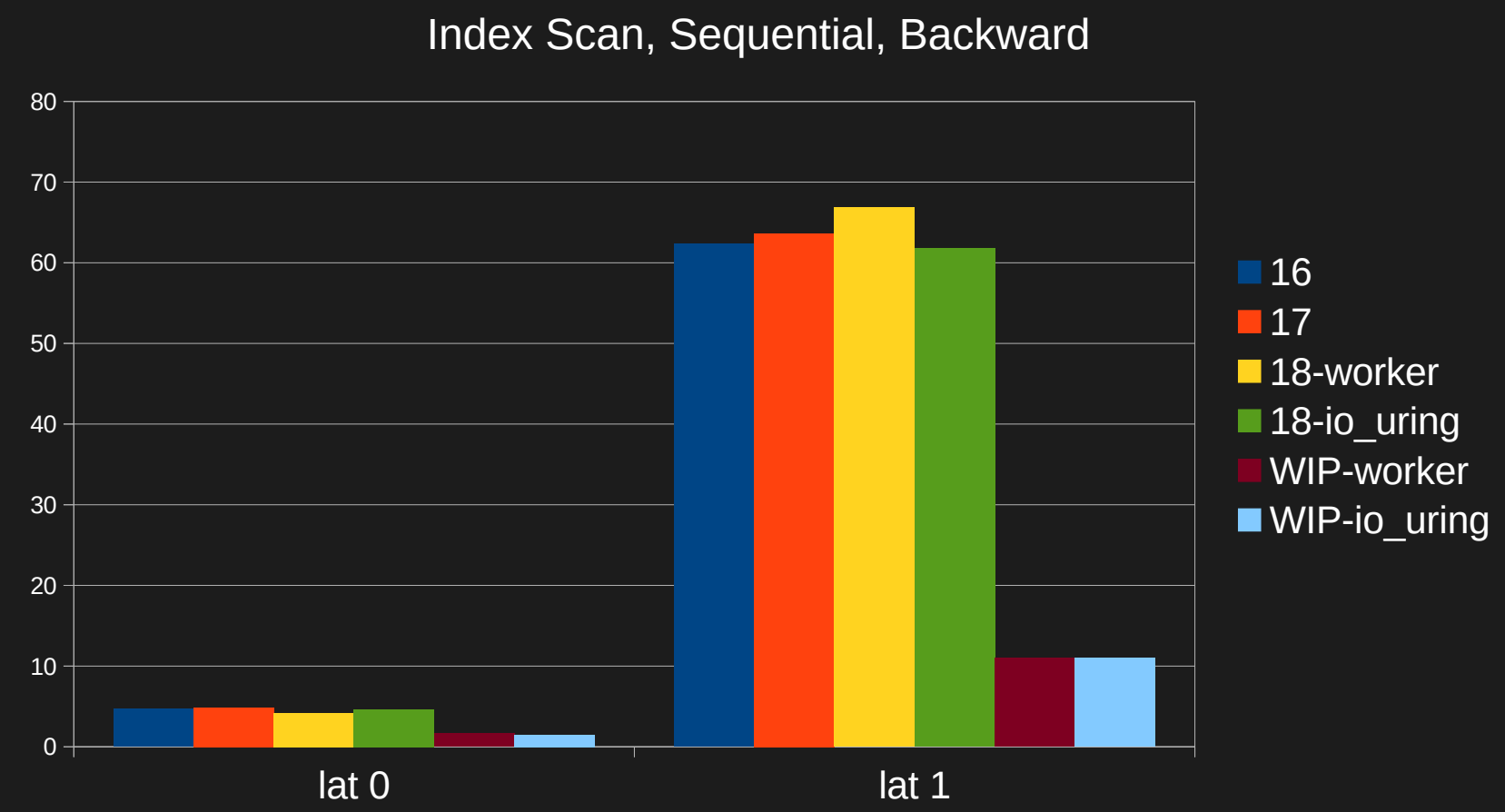
# 19? 20?: Index Readahead




- No kernel readahead!
- Substantial gain w & w/o added latency

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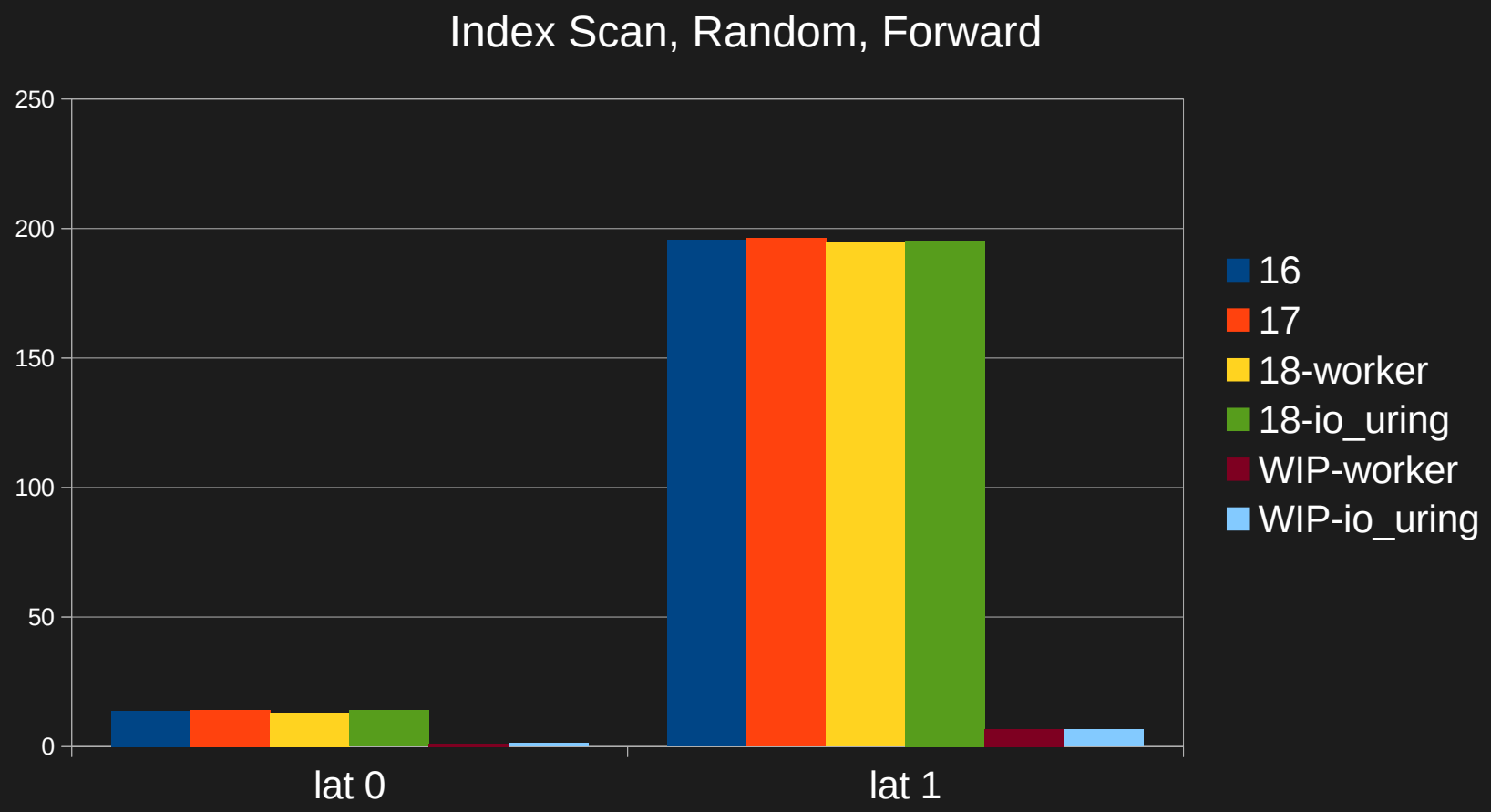
# 19? 20?: Index Readahead




- No kernel readahead
- Small IOs
- Huge wins

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# 19?: AIO writes in bgwriter & checkpointer

- Infrastructure for Buffered AIO writes required
- 2-3x checkpoint speed for sequential data
- bigger for large amounts of random data

Microsoft

# 19?: AIO for COPY & VACUUM

- Infrastructure for Buffered AIO writes required
- 2-4x speedup observable
- Bottleneck often elsewhere
  - WAL (COPY & VACUUM)
  - index reads during insertion (COPY)

Microsoft

# 20, 21?: AIO for WAL writes

- Hard
- Huge wins possible
- Helpful for
  - Bulk load, VACUUM
  - Concurrent OLTP workloads
- Not helpful for
  - low concurrency OLTP

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# Future AIO Users

- Recovery Readahead
  - crucial for working without full-page-writes / RWF_ATOMIC
- alter database set tablespace
- create database reads (strategy file_copy) & writes
- fsyncing files at end of checkpoint
- unlinking lots of files
- ...

Microsoft

# Other Future Work

- Other IO methods
  - Windows IOCP or io_uring
  - FreeBSD (+others?) posix_aio
- Optimize existing code
  - auto-tune number of workers
  - registered buffers for io_uring
- Integrate async network IO

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